Hsiang-Wei Huang

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Summary

Applied Scientist II at **Amazon Fauna Robotics**, working on the robot's human perception stack, including person detection, pose estimation, multi-object tracking, active speaker detection, gesture recognition, and face recognition, and developing open-vocabulary detection and tracking for robotic manipulation. Ph.D. in Electrical and Computer Engineering from the University of Washington, advised by Prof. Jenq-Neng Hwang, with a dissertation on *Building Multimodal AI Systems for Perceiving and Reasoning in 3D Space*. Research spans **Multimodal AI**, **3D Scene Understanding**, **Spatial Intelligence**, and **Vision-Language Models**.

Education

Ph.D. in Electrical and Computer Engineering — Seattle, WA
University of Washington — Mar. 2023 - Jun. 2026

- **Dissertation:** Building Multimodal AI Systems for Perceiving and Reasoning in 3D Space
- **Advisor:** Jenq-Neng Hwang

M.S. in Electrical and Computer Engineering — Seattle, WA
University of Washington — Sept. 2021 - Dec. 2022

Bachelor of Science — Hsinchu, Taiwan
National Chiao Tung University — Sept. 2017 - Dec. 2020

Experience

Amazon, Fauna Robotics — Bellevue, WA
Applied Scientist II — Aug. 2026 - Present

- **Human Perception Stack.** Work on the robot's human perception pipeline spanning person detection, pose estimation, and multi-object tracking, deployed as a core runtime component of the robot platform.
- **VLM-in-the-Loop Auto-Labeling.** Designed a vision-language-model-in-the-loop pipeline that automatically generates pseudo-ground-truth person tracks, improving tracking accuracy by **9.7%** HOTA over the production tracker.
- **Active Speaker Detection.** Improved the robot's active speaker detection system by **8.2%** mAP. Designed a streaming evaluation protocol that quantifies the offline-to-online performance gap for real-time robot deployment.
- **Gesture Recognition.** Developed a real-time skeleton-based gesture recognition system that lets the robot recognize human gestures such as waving, hand signs, head shaking, and clapping for natural human-robot interaction.
- **Face Recognition.** Integrated face recognition into the robot's perception stack as an identity layer above the person tracker, keeping person identities consistent across long interactions.
- **Open-Vocabulary Perception.** Drove improvements to object detection and tracking for robotic manipulation tasks.

Amazon, Device CoRo — Bellevue, WA
Applied Scientist Intern — Jun. 2025 - Sept. 2025

- **Efficient 3D Scene Understanding.** Developed efficient 3D scene understanding vision-language and vision foundation models for video question answering.
- **Publications.** First-author paper *Memory Tree Guided Key Frame Querying for Efficient 3D Question Answering* accepted to **ECCV 2026**; co-authored *DiffGround: Semantic Difference-Guided Zero-Shot Video Question Grounding* accepted to **ACCV 2026**.

Amazon, Device CoRo — Bellevue, WA
Applied Scientist Intern — Jun. 2024 - Sept. 2024

- **3D Scene Understanding LLMs.** Researched 3D scene understanding large language models and open-vocabulary indoor 3D instance segmentation.
- **Publications.** First-author paper *Zero-shot 3D Question Answering via Voxel-based Dynamic Token Compression* accepted to **CVPR 2025**; co-authored *Details Matter for Indoor Open-vocabulary 3D Instance Segmentation* accepted to **ICCV 2025**.

Chimei Automotive Taiwan

AI Research Intern Feb. 2023 - Mar. 2023

- **Forward Collision Warning.** Developed a real-time multi-class Forward Collision Warning System for trucks and construction vehicles.
- **Lane Departure Warning.** Improved Lane Departure Warning System accuracy by **30%** under poor visibility conditions using an autoencoder-based approach.

Selected Publications

- **Memory Tree Guided Key Frame Querying for Efficient 3D Question Answering**, *European Conf. on Computer Vision (ECCV)*, 2026.
  H.-W. Huang, F.-C. Chen, L.-W. Tsao, C.-H. Lee, C.-C. Su, L. Xia, R. Peng, J.-N. Hwang, M. Sun, C.-H. Kuo.
- **Beyond Direct Answers: Camera Motion Grounded Training and Evaluation for Vision-Language Models**, *British Machine Vision Conference (BMVC)*, 2026 — **Oral**.
  H.-W. Huang, J. Lu, K.-M. Chen, J. Shangguan, C.-Y. Yang, J.-N. Hwang.
- **SpatialAgent: Spatial Question Answering with LLM Agent and Perception Models**, *IEEE Int. Conf. on Multimedia Information Processing and Retrieval (MIPR)*, 2026 — **Best Student Paper Award**.
  H.-W. Huang, J. Lu, J. Shangguan, J.-H. Cheng, K.-M. Chen, C.-Y. Yang, B. Alattar, Y.-R. Lin, P. Kim, S. Kim, K. Kim, C.-I Huang, J.-N. Hwang.
- **Zero-shot 3D Question Answering via Voxel-based Dynamic Token Compression**, *IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2025.
  H.-W. Huang, F.-C. Chen, W. Chai, C.-C. Su, L. Xia, S. Jung, C.-Y. Yang, J.-N. Hwang, M. Sun, C.-H. Kuo.
- **ToSA: Token Merging with Spatial Awareness**, *IEEE/RSJ Int. Conf. on Intelligent Robots and Systems (IROS)*, 2025 — **Oral**.
  H.-W. Huang, W. Chai, K.-M. Chen, C.-Y. Yang, J.-N. Hwang.
- **MambaMOT: State-Space Model as Motion Predictor for Multi-Object Tracking**, *IEEE Int. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*, 2025 — **Oral**.
  H.-W. Huang, C.-Y. Yang, W. Chai, Z. Jiang, J.-N. Hwang.
- **Warehouse Spatial Question Answering with LLM Agent**, *ICCV AI City Challenge Workshop*, 2025 — **1st Place, Track 3: Warehouse Spatial Intelligence**.
  H.-W. Huang, J.-H. Cheng, K.-M. Chen, C.-Y. Yang, B. Alattar, Y.-R. Lin, P. Kim, S. Kim, K. Kim, C.-I Huang, J.-N. Hwang.
- **3D Visual Grounding with Reasoning LLM**, *CVPR 3D-LLM & VLA Workshop*, 2025.
  H.-W. Huang, K.-M. Chen, W. Chai, C.-Y. Yang, J.-N. Hwang.
- **ToddlerAct: A Toddler Action Recognition Dataset for Gross Motor Development Assessment**, *ECCV ABAW Workshop*, 2024.
  H.-W. Huang, J. Sun, C.-Y. Yang, Z. Jiang, J.-N. Hwang.
- **Iterative Scale-up Expansion and Deep Features Association for Multi-Object Tracking in Sports**, *WACV RWS Workshop*, 2024.
  H.-W. Huang, C.-Y. Yang, J. Sun, P.-K. Kim, K.-J. Kim, C.-I Huang, J.-N. Hwang.
- **Enhancing Multi-Camera People Tracking with Anchor-Guided Clustering and ID Re-Assignment**, *CVPR AI City Challenge Workshop*, 2023 — **1st Place**.
  H.-W. Huang, C.-Y. Yang, Z. Jiang, P.-K. Kim, K. Lee, K. Kim, S. Ramkumar, C. Mullapudi, I.-S. Jang, C.-I Huang, J.-N. Hwang.
- **Observation Centric and Central Distance Recovery for Athlete Tracking**, *WACV CV4WS Workshop*, 2023.
  H.-W. Huang, C.-Y. Yang, S. Ramkumar, C.-I Huang, J.-N. Hwang, P.-K. Kim, K. Lee, K. Kim.
- **SAMURAI: Adapting SAM 2 for Visual Object Tracking with Motion Cues**, *IEEE Transactions on Image Processing (TIP)*, 2026.
  C.-Y. Yang, H.-W. Huang, W. Chai, Z. Jiang, J.-N. Hwang.
- **CLEP: Contrastive Language-Pose Pretraining**, *IEEE/CVF Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2026.
  S. Jia, H. Wang, H.-W. Huang, Z. An, J.-N. Hwang, H. Zhang, L. Li.
- **Extending Segment Anything Model 2 to Multi-Object Tracking by Optimizing Hierarchical Trajectory Memory**, *CVPR Findings*, 2026.
  C.-Y. Yang, H.-W. Huang, K.-M. Chen, K. Li, J.-N. Hwang.
- **DiffGround: Semantic Difference-Guided Zero-Shot Video Question Grounding**, *Asian Conf. on Computer Vision (ACCV)*, 2026.
  C.-H. Lee, A. Salimi, H.-W. Huang, C.-C. Su, M. Sun, C.-H. Kuo.
- **Details Matter for Indoor Open-vocabulary 3D Instance Segmentation**, *Int. Conf. on Computer Vision (ICCV)*, 2025.
  S. Jung, J. Zheng, K. Zhang, N. Qiao, A. Y. C. Chen, L. Xia, C. Liu, Y. Sun, X. Zeng, H.-W. Huang, B. Boots, M. Sun, C.-H. Kuo.
- **RT-Pose: A 4D Radar Tensor-based 3D Human Pose Estimation and Localization Benchmark**, *European Conf. on Computer Vision (ECCV)*, 2025.
  Y.-H. Ho, J.-H. Cheng, S. Y. Kuan, Z. Jiang, W. Chai, H.-W. Huang, C.-L. Lin, J.-N. Hwang.
- **DAMURA: A Depth-Aware Robust Multi-Object Tracker for Crowded Scenes by Re-Prioritizing Association Order**, *IEEE Int. Conf. on Advanced Video and Signal-Based Surveillance (AVSS)*, 2025.
  C.-Y. Yang, H.-W. Huang, K.-M. Chen, K. Li, F. Wallace, C.-I Huang, J.-N. Hwang.

- **A Density-Guided Temporal Attention Transformer for Indiscernible Object Counting in Underwater Videos**, *IEEE Int. Conf. on Acoustics, Speech and Signal Processing (ICASSP)*, 2024.
  C.-Y. Yang, H.-W. Huang, Z. Jiang, H. Wang, F. Wallace, J.-N. Hwang.
- **Boosting Online 3D Multi-Object Tracking through Camera-Radar Cross Check**, *IEEE Intelligent Vehicles Symposium (IV)*, 2024.
  S.-Y. Kuan, J.-H. Cheng, H.-W. Huang, W. Chai, C.-Y. Yang, H. Latapie, G. Liu, B.-F. Wu, J.-N. Hwang.
- **Adapting SAM2 for Visual Object Tracking**, *ICPR MMVPR Workshop*, 2024 — **1st Place, Multi-Modal Tracking Challenge**.
  C.-Y. Yang, H.-W. Huang, P.-K. Kim, C.-K. Kuo, J.-W. Chang, J.-N. Hwang.
- **Sea You Later: Metadata-Guided Long-Term Re-Identification for UAV-Based Multi-Object Tracking**, *WACV MaCVi Workshop*, 2024 — **1st Place**.
  C.-Y. Yang, H.-W. Huang, Z. Jiang, H.-C. Kuo, J. Mei, J.-N. Hwang.
- **An Online Approach and Evaluation Method for Tracking People Across Cameras in Extremely Long Video Sequences**, *CVPR AI City Challenge Workshop*, 2024.
  C.-Y. Yang, H.-W. Huang, P.-K. Kim, Z. Jiang, J.-N. Hwang.
- **GTA: Global Tracklet Association for Multi-Object Tracking in Sports**, *ACCV MLCSA Workshop*, 2024.
  J. Sun, H.-W. Huang, C.-Y. Yang, Z. Jiang, J.-N. Hwang.

Honors & Awards

Best Student Paper Award, IEEE MIPR 2026
- Awarded for *SpatialAgent: Spatial Question Answering with LLM Agent and Perception Models*.

CVPR Outstanding Reviewer 2026
- Recognized as an Outstanding Reviewer at CVPR 2026.

ICML Silver Reviewer 2026
- Recognized among the top reviewers for ICML 2026 based on area chair ratings of submitted reviews.

Arthur Y.C. Chiao Global Exchange Ph.D. Fellowship 2025
- Exchange fellowship to National Yang Ming Chiao Tung University (NYCU), Taiwan.

Taiwan Future Tech Award 2024
- Collaborated with Taiwan's National Center for High-performance Computing on a traffic monitoring framework.
- Awarded for *Digital City — Smart Traffic Congestion Prediction and Police Service Support System*.

Competition Awards

1st Place, ECCV AI City Challenge — Egocentric Dashcam Pedestrian Intent VQA 2026
- Ranked 1st on the out-of-domain egocentric dashcam pedestrian intent VQA leaderboard with **70.64%** accuracy.

1st Place, CVPR Thermal Pedestrian Multiple Object Tracking Challenge 2026
- Ranked 1st in the 2nd Thermal Pedestrian Multiple Object Tracking Challenge at CVPR.

1st Place, ICCV AI City Challenge — Warehouse Spatial Question Answering 2025
- Ranked 1st with **95.8%** question answering accuracy; solution presented at the ICCV AI City Challenge workshop.

2nd Place, WACV SkiTB Visual Tracking Challenge 2025
- Mentored an undergraduate student to a 2nd place finish with an **87.0 F1** score.

1st Place, ICPR Multi-modal Tracking Challenge 2024
- Ranked 1st with **89.4 AUC** among 53 participants; solution presented at the ICPR MMVPR workshop.

1st Place, WACV MaCVi Challenge — UAV-based Multi-Object Tracking and Re-Identification 2024
- Proposed a metadata-aided re-identification method for long-term tracking; best performance with **69.5 HOTA**.

1st Place, WACV MaCVi Challenge — USV-based Multi-Object Tracking 2024
- Proposed a model ensemble method achieving the highest HOTA in the WACV Maritime Computer Vision Challenge.

1st Place, CVPR AI City Challenge — Multi-Camera People Tracking 2023
- Proposed a robust anchor-guided clustering method for multi-camera people tracking and re-identification.
- Achieved best performance with an **IDF1 of 95.36** on the Track 1 public test set.

3rd Place, ECCV DeeperAction Challenge — SportsMOT Multi-actor Tracking 2022
- Achieved over **73.9 HOTA** on sports player tracking across three sports scenes; presented at the ECCV DeeperAction workshop.